Epidemiology - Week 1 - Introduction to Epidemiology

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What is Epidemiology?

It is the study of the **frequency and distribution of health-related states or events in specific populations**, including the study of determinants influencing such states, and the application of this study to control health problems.

Basically, studying the frequency or distribution of health-related outcomes within specific populations alongside its causation and correlation with the goal of translating the insight into public health actions.

Role of Epidemiology

- **DESCRIBE** - the patterns of health outcome and disease within populations
- **INTERPRET** - the differences
- **APPLY** - the results to public health practice
- **EVALUATE** - the effect of the application / intervention

Cases and Populations

Quantification needs a clear case definition and a defined study population.

- **CASE**: A set of standard criteria to decide whether a person has a specific disease, syndrome, or health condition
- **STUDY POPULATION**: The population who could become or have become cases, AKA **population at risk**

Exposure and Outcome

- **EXPOSURE**: Risk factor or determinant -¿ Any factor that may influence the outcome
- **OUTCOME**: Disease, event, or health related state of interest, or risk factor / precursor to a disease

Types of Epidemiological Investigation / Study Design

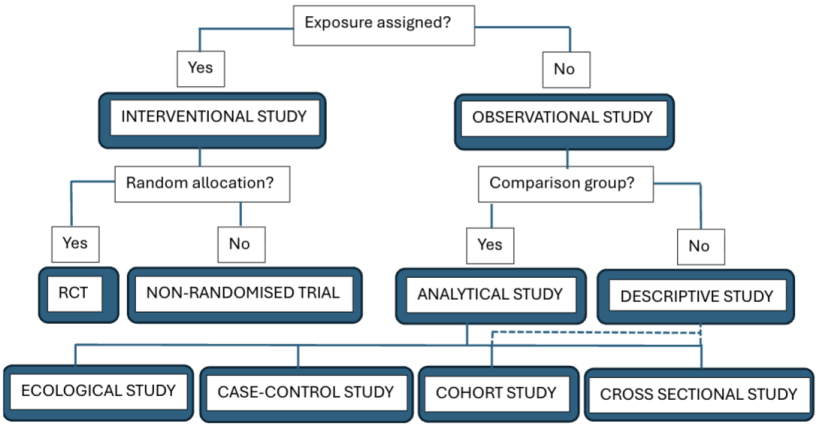


Figure 1: Types of epidemiological investigation

Observational Epidemiology

This type of epidemiology is based on examining the distribution and determinants of an outcome **without any attempt to influence it**, and is split into two further subtypes.

- **DESCRIPTIVE**: Designed to describe the distribution of an exposure or outcome without attempting to explain (or influence) the distribution -> *Who, what, were, when?*
- **ANALYTICAL**: Designed to examine **association** with the aim of identifying causes for an outcome -> *Why?*

Outbreaks and Epidemics

Some definitions of disease occurrence

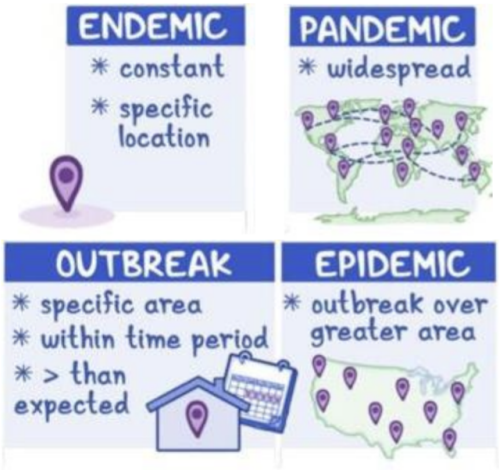


Figure 2: Disease occurrence types

- **ENDEMIC**: Common at baseline level in a given area / population
- **OUTBREAK**:
  - Excess of cases of a disease compared with *expected* number
  - Limited to a local increase in disease (often infectious) incidence; e.g. Town, village, institution
- **EPIDEMIC**:
  - Excess of cases of a disease compared with *expected* number
  - Generalised, cases distributed across wider area / time period
- **PANDEMIC**: Affects large numbers of people and crosses international boundaries

Examples of Types of Epidemics

Type	Description	Example
Common-source	Individuals are exposed to the same source of infection / hazard	Food contamination at wedding
Propagated	Disease spreads from person-to-person	Measles outbreak at school
Mixed	A combination of common-source followed with propagation	Cholera outbreak that started from a contaminated water source followed with household transmission

Table 1: Example of types of epidemics

Clusters

An aggregation of some relatively uncommon disease or event which can include infectious and non-communicable conditions. Often unknown / multiple aetiological factors with low exposure level over number of years.

Pandemics

A pandemic is an epidemic that *spreads across multiple countries or continents* and *affects large numbers of people, involving a novel pathogen or population susceptibility.* Examples,

- 1918-1919 Influenza pandemic
- Covid-19

Strategy against Pandemics

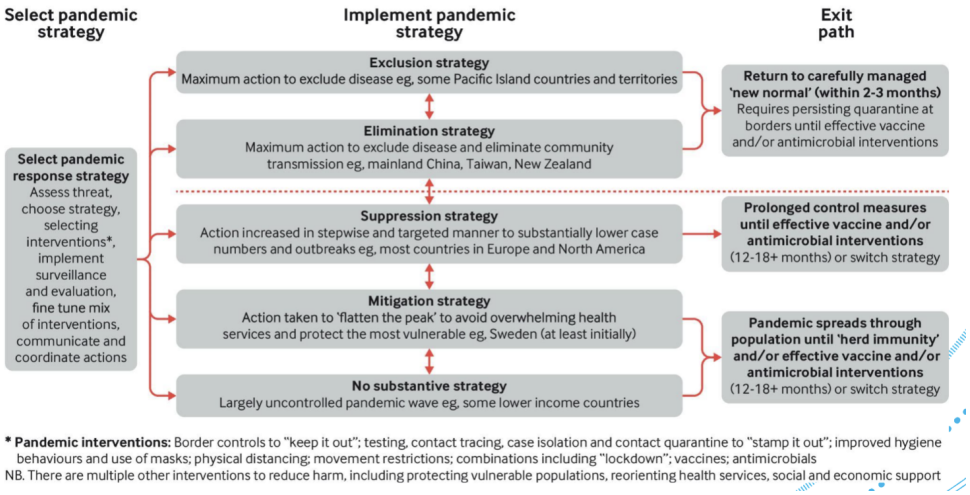


Figure 3: Pandemic strategy choices

Elimination vs Mitigation

Eliminate literally just means trying to eliminate / stop the community transmission of the disease, while mitigation aims to slow the transmission down to a manageable level so healthcare systems don't get overwhelmed.